

International University of Valencia (VIU)

Master's Degree in Data Science and Big Data
Advanced Statistics



Programming in RStudio

Activity 2 - Time series analysis

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1. Introduction

1.1 Context and motivation

Nitrogen dioxide (NO₂) originates primarily from combustion processes, particularly road traffic, and prolonged exposure is associated with adverse effects on the respiratory system. The Swiss Air Protection Ordinance (*Luftreinhalte-Verordnung*, LRV) establishes emission limit values that the authorities must continuously monitor.

The city of Zurich has been measuring air quality since 1983 and publishing the results as open data. These data series are ideal for temporal analysis, as they combine an underlying trend associated with vehicle fleet renewal, annual seasonality resulting from winter temperature inversions that limit pollutant dispersion, and an irregular meteorological component. Having a model capable of predicting its evolution allows us to assess whether the measures taken are maintaining the downward trend and to forecast the months of greatest risk.

1.2 Objectives of the analysis

1. Statistically describe the series and quantify its trend, seasonality, and irregularity components.
2. Determine the appropriate type of decomposition based on the relationship between the level of the series and its variability.
3. Identify the model orders from the autocorrelation analysis.
4. Estimate a SARIMA model, compare it with alternative methods, and validate its predictive capacity on data not used in the fitting.
5. Verify the model assumptions and predict the evolution of NO₂ for 2026.

The analysis was developed in R 4.6.1 on RStudio, using the *forecast*, *tseries*, *dplyr*, *lubridate*, and *ggplot2* packages. The complete code is included in Appendix A.

2. Data Description

2.1 Origin and preparation

The data comes from the dataset *Tglich aktualisierte Luftqualittsmessungen, seit 1983*, published by the Zurich City Health and Environment Department (UGZ) under a CC0 license. The format is long, with seven variables: date, station, pollutant, interval, unit, value, and treatment status. The annual files for the period 2010–2025 were downloaded, totaling 179,450 records.

```
> str(prueba)
'data.frame': 13176 obs. of 7 variables:
 $ Datum : chr "2024-01-01T00:00+0100" "2024-01-01T00:00+0100" "2024-01-01T00:00+0100" "2024-01-01T00:00+0100" ...
 $ Standort : chr "Zch_Stampfenbachstrasse" "Zch_Stampfenbachstrasse" "Zch_Stampfenbachstrasse" "Zch_Stampfenbachstrasse" ...
 $ Parameter: chr "SO2" "CO" "NOx" "NO" ...
 $ Intervall: chr "d1" "d1" "d1" "d1" ...
 $ Einheit : chr "µg/m3" "mg/m3" "ppb" "µg/m3" ...
 $ Wert : num 1.06 0.19 6.23 1.55 9.53 ...
 $ Status : chr "bereinigt" "bereinigt" "bereinigt" "bereinigt" ...
> head(prueba)
      Datum      Standort Parameter Intervall Einheit
1 2024-01-01T00:00+0100 Zch_Stampfenbachstrasse SO2 d1 µg/m3
2 2024-01-01T00:00+0100 Zch_Stampfenbachstrasse CO d1 mg/m3
3 2024-01-01T00:00+0100 Zch_Stampfenbachstrasse NOx d1 ppb
4 2024-01-01T00:00+0100 Zch_Stampfenbachstrasse NO d1 µg/m3
5 2024-01-01T00:00+0100 Zch_Stampfenbachstrasse NO2 d1 µg/m3
6 2024-01-01T00:00+0100 Zch_Stampfenbachstrasse O3 d1 µg/m3
      Wert      Status
1 1.06 bereinigt
2 0.19 bereinigt
3 6.23 bereinigt
4 1.55 bereinigt
5 9.53 bereinigt
6 56.41 bereinigt
```

Figure 1. Data set structure.

Of the four available stations, Stampfenbachstrasse was selected, located in a representative urban environment, and of the eleven parameters, NO₂ in µg/m³ was chosen. The filtering process yielded 5,844 daily observations that cover the period from January 1, 2010, to December 31, 2025, without duplicate dates and with all records in a clean state. Only 19 values are missing, representing 0.33% distributed across ten different months.

```
> nrow(no2)
[1] 5844
> range(no2$fecha)
[1] "2010-01-01" "2025-12-31"
> sum(is.na(no2$Wert))
[1] 19
> sum(duplicated(no2$fecha))
[1] 0
> table(no2$Status)

bereinigt
5844
> summary(no2$Wert)
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
      2.36  16.80   25.27   27.71  36.10   96.51    19
```

Figure 2. Quality checks of the daily series.

The data were aggregated monthly by averaging the daily values. This avoids working with a frequency of 365 observations per cycle and filters out short-term meteorological variability, while preserving annual seasonality. The resulting series consists of 192 observations and is obtained using the code specified in Annex A. Missing values were excluded from the calculation without imputation; the month with the least coverage retains 25 valid days.

2.2 Exploratory Analysis

The series shows a mean of $27.75 \mu\text{g}/\text{m}^3$ and a median of 26.52 , with a range between 9.36 and 55.42 . The standard deviation of 9.76 yields a coefficient of variation of 35.2% . Annual averages decrease from $36.25 \mu\text{g}/\text{m}^3$ in 2010 to 18.22 in 2025, a reduction of 49.7% .

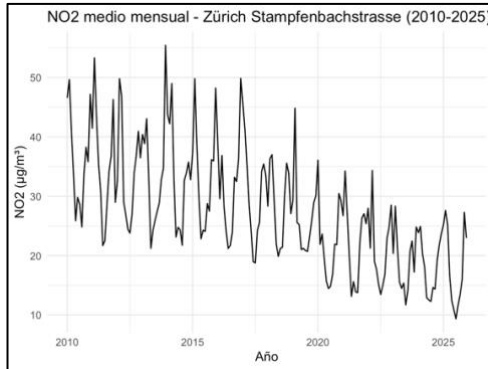


Figure 3. Monthly series of NO₂ (2010-2025).

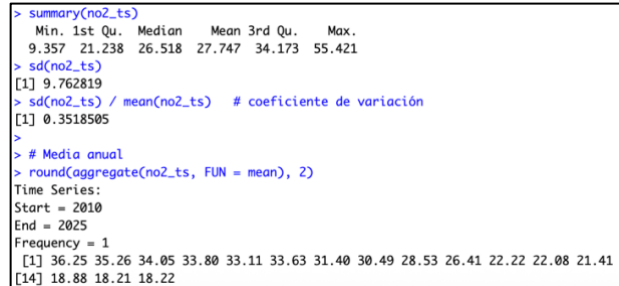


Figure 4. Descriptive statistics and annual averages.

The graph reveals three features: a decreasing trend, regular annual seasonality, and, notably, a progressive reduction in the amplitude of oscillations as the level decreases. Maximum values are concentrated between December and March, with February at the top (median of $37.5 \mu\text{g}/\text{m}^3$), and minimum values between May and August, with medians close to 21 . December is the month with the greatest dispersion, which is consistent with the episodic nature of temperature inversions. The superimposed seasonal graph shows that the shape of the cycle remains stable, while the curves for recent years are lower and have a flatter profile.

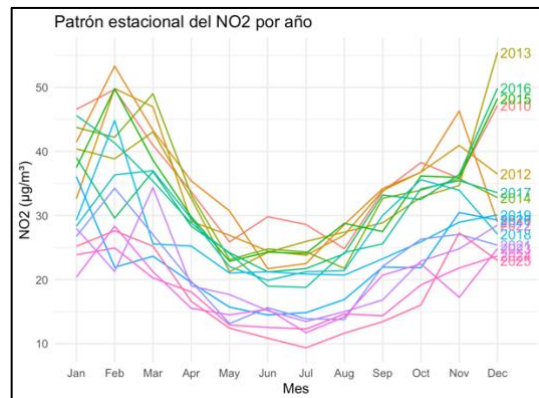
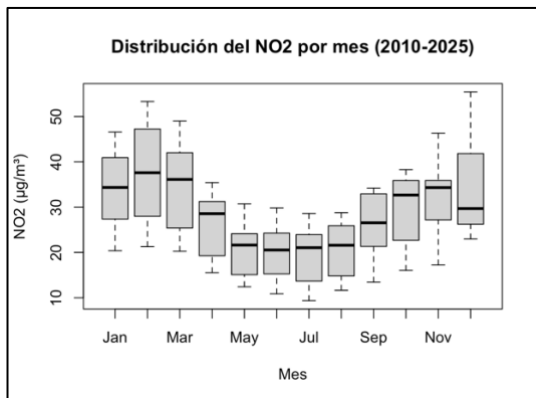


Figure 5. (a) Distribution of NO₂ by month. (b) Seasonal pattern by year.

2.3 Decomposition of the series

The correlation between the annual mean and the annual standard deviation is 0.883 : variability decreases in parallel with the level, going from values around 9 in the first period to values close to 5 in recent years. Seasonality, therefore, acts as a percentage of the level and not as a fixed quantity, which justifies a multiplicative decomposition.

```
> medias_anuales <- as.numeric(aggregate(no2_ts, FUN = mean))
> sd_anuales <- as.numeric(aggregate(no2_ts, FUN = sd))
> round(sd_anuales, 2)
[1] 8.44 9.56 8.53 9.54 8.69 9.03 8.39 8.50 6.31 6.73 6.77 6.82 6.46 4.93 4.75 6.94
> cor(medias_anuales, sd_anuales)
[1] 0.8826588
```

```
> desc_mult <- decompose(no2_ts, type = "multiplicative")
>
> round(desc_mult$figure, 3) # índices estacionales
[1] 1.206 1.313 1.224 0.921 0.734 0.697 0.695 0.761 0.944 1.090 1.163 1.251
> range(desc_mult$trend, na.rm = TRUE) # rango de la tendencia
[1] 17.95605 36.66490
> sd(desc_mult$random, na.rm = TRUE)
[1] 0.1128365
```

Figure 6. (a) Relationship between level and annual variability. (b) Estimated components

The trend shows a decrease from 36.66 to 17.96 $\mu\text{g}/\text{m}^3$ and can be divided into three phases: a moderate decline until 2017, a sharp drop between 2017 and 2020, and stabilization around 18-19 $\mu\text{g}/\text{m}^3$ since 2022. Seasonal indices place the maximum in February (1.313, 31% above the trend) and the minimum in July (0.695, 30% below), meaning that the winter level is almost double the summer level. The irregular component has a standard deviation of 0.113 and fluctuates around one with no apparent pattern.

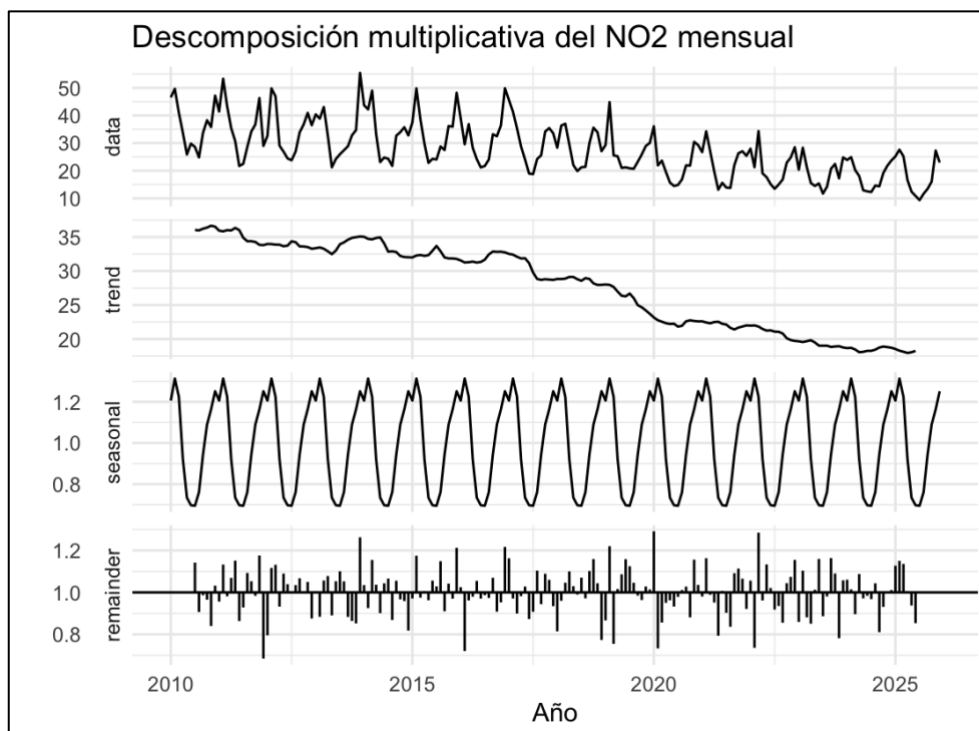


Figure 7. Multiplicative decomposition of the series.

3. Analysis

3.1 Transformation and stationarity

The multiplicative nature of the series suggests applying a logarithmic transformation before modeling it, as this makes the relationship between components additive and stabilizes the variance. The effect is noticeable: the amplitude of the oscillations becomes comparable at the beginning and end of the period.

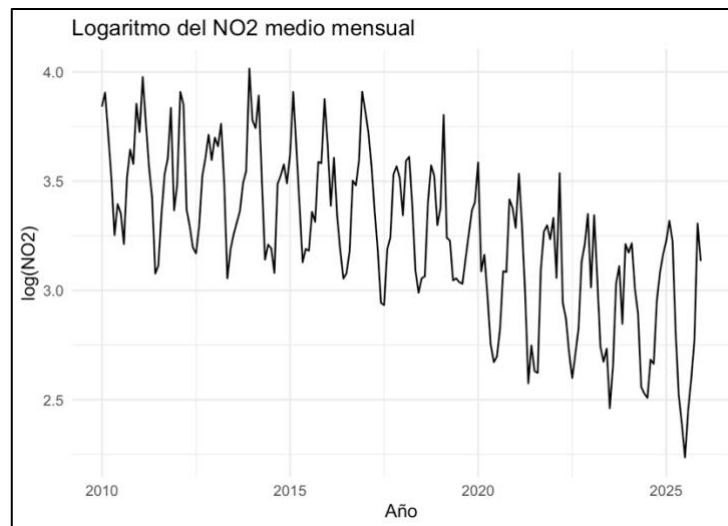


Figure 8. Logarithmically transformed series.

Two complementary tests were applied to the logarithmic series. The augmented Dickey-Fuller (ADF) test, whose null hypothesis is the existence of a unit root, yields a statistic of -9.43 with $p < 0.01$, so it is rejected. The KPSS test, whose null hypothesis is stationarity at the level, yields a statistic of 2.51 with $p < 0.01$, so it is also rejected.

This apparent contradiction has a methodological explanation. The ADF incorporates sufficient lags to absorb the seasonal structure and concludes that there is no unit root, while the KPSS detects what the graph clearly shows: the mean is not constant, but rather decreases from 36 to $18 \mu\text{g}/\text{m}^3$ over the period. It is precisely this changing mean that needs to be corrected. The `ndiffs` and `nsdiffs` functions confirm the diagnosis and recommend a regular difference ($d = 1$) and a seasonal difference ($D = 1$).

```

Augmented Dickey-Fuller Test

data: log_no2
Dickey-Fuller = -9.4267, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In adf.test(log_no2) : p-value smaller than printed p-value

> kpss.test(log_no2, null = "Level")

KPSS Test for Level Stationarity

data: log_no2
KPSS Level = 2.5134, Truncation lag parameter = 4, p-value = 0.01

Warning message:
In kpss.test(log_no2, null = "Level") :
  p-value smaller than printed p-value

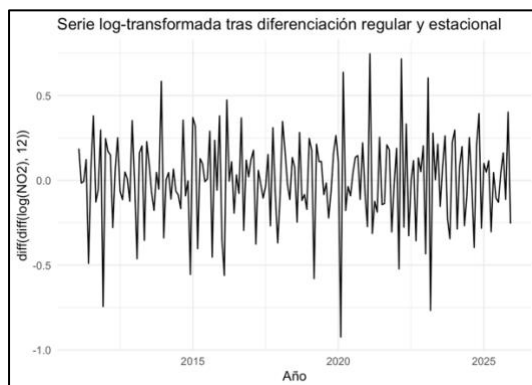
>
> # 3. Número de diferencias recomendadas
> ndiffs(log_no2)
[1] 1
> nsdiffs(log_no2)
[1] 1

```

Figure 9. Stationarity tests on the logarithmic series.

3.2 Differentiation

First, the 12th-order seasonal difference was applied, which eliminates seasonality by subtracting the value of the same month of the previous year from each month's value, followed by the regular difference, which eliminates the residual trend. The resulting series consists of 179 observations, after discounting the thirteen that are lost in the process.



```

> # Comprobación de estacionariedad
> adf.test(log_diff)

Augmented Dickey-Fuller Test

data: log_diff
Dickey-Fuller = -9.1363, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In adf.test(log_diff) : p-value smaller than printed p-value

> kpss.test(log_diff, null = "Level")

KPSS Test for Level Stationarity

data: log_diff
KPSS Level = 0.013875, Truncation lag parameter = 4, p-value = 0.1

Warning message:
In kpss.test(log_diff, null = "Level") :
  p-value greater than printed p-value

>
> length(log_diff) # 192 - 12 - 1 = 179 observaciones
[1] 179

```

Figure 10. (a) Series after double differentiation. (b) Stationarity contrasts.

The differencing series oscillates around zero with an approximately constant amplitude and no appreciable trend or seasonal pattern. Both tests now agree on the diagnosis: the ADF rejects the unit root ($p < 0.01$) and the KPSS does not reject stationarity ($p > 0.1$). The series is suitable for the identification phase.

3.3 Model Identification

With 179 observations, the significance threshold for autocorrelations is ± 0.147 . The simple autocorrelation function (ACF) has a value of -0.467 at lag 1, after which the coefficients fall below the threshold, and a value of -0.549 at lag 12, while lag 24 ceases to be significant. The partial autocorrelation function (PACF) shows a gradual decay in the first lags (-0.467 , -0.392 , and -0.332) and also in the seasonal lags.

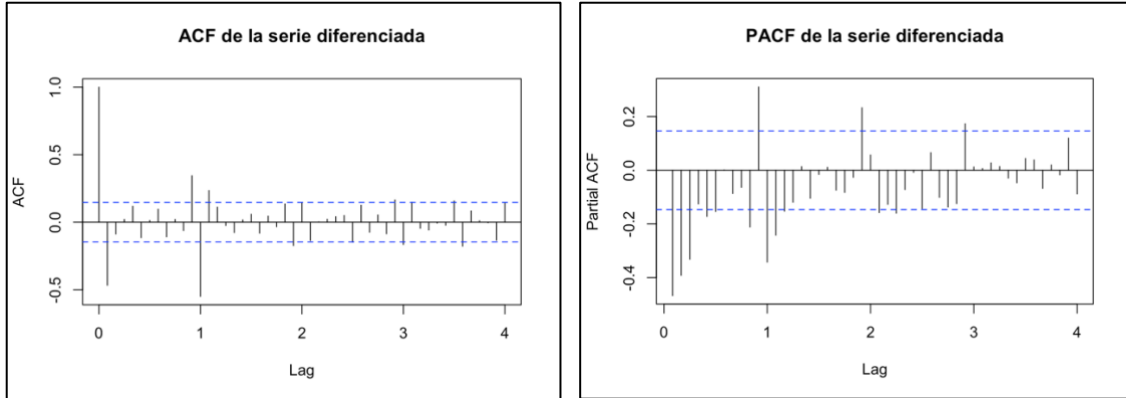


Figure 11. (a) ACF and (b) PACF of the differentiated series.

This combination, a cut in the ACF accompanied by a decay in the PACF, is characteristic of a moving average process in both the regular and seasonal parts. The significant coefficients in delays 11 and 13 correspond to the expected cross-effect between both parts and do not constitute independent components. The candidate model is, therefore, an $\text{ARIMA}(0,1,1)(0,1,1)$ [12], known as the airline model due to its original formulation in air traffic analysis.

3.4 Estimation and selection

To evaluate the predictive capacity, the series was divided into a training set with the 168 months between 2010 and 2023 and a test set with the 24 months of 2024 and 2025. Along with the identified model, two alternatives were estimated that incorporate an additional parameter in the regular part.

	modelo	AIC	BIC
1	$(0,1,1)(0,1,1)$	-146.51	-137.38
2	$(1,1,1)(0,1,1)$	-144.52	-132.34
3	$(0,1,2)(0,1,1)$	-144.52	-132.34

Figure 12. Comparison of the candidate models.

The airline model obtains the best values in both information criteria ($\text{AIC} = -146.51$; $\text{BIC} = -137.38$) compared to the alternatives ($\text{AIC} = -144.52$; $\text{BIC} = -132.34$), so the incorporation of an $\text{AR}(1)$ or $\text{MA}(2)$ term does not compensate for the additional parameter. Applying the parsimony criterion, the model identified in the previous phase is selected.

```

> mod_airline
Series: train
ARIMA(0,1,1)(0,1,1)[12]

Coefficients:
            ma1      sma1
      -0.8958  -0.9999
s.e.   0.0334   0.0954

sigma^2 = 0.01783: log likelihood = 76.26
AIC=-146.51  AICc=-146.35  BIC=-137.38

```

Figure 13. Estimated ARIMA(0,1,1)(0,1,1)[12] model.

Both coefficients are highly significant: the regular moving average term is -0.896 , with a standard error of 0.033 , and the seasonal term is -1.000 , with a standard error of 0.095 . The residual variance is 0.0178 , equivalent to a standard error of approximately 13% on the original scale. The value of the seasonal coefficient, located at the limit of invertibility, indicates that the seasonal component is practically deterministic, a common circumstance in series with a very stable annual pattern.

3.5 Comparison with alternative methods

The resulting model was tested against two different approaches: automatic selection using `auto.arima`, configured to exhaustively explore the model space, and the Holt-Winters exponential smoothing method in its additive version, equivalent to a multiplicative one on the original scale since the series is in logarithms.

```

Series: train
ARIMA(1,0,1)(2,1,0)[12] with drift

Coefficients:
      ar1      ma1      sar1      sar2      drift
    0.9685 -0.9020 -0.7464 -0.3868 -0.0041
s.e.  0.0308  0.0505  0.0785  0.0778  0.0014

sigma^2 = 0.02405: log likelihood = 68.07
AIC=-124.15  AICc=-123.58  BIC=-105.85

```

```

alpha
0.0579275
beta
0.03910589
gamma
0.1957644
> mod_hw$coefficients[1:2]
      a      b
2.915096491 -0.006044645

```

Figure 14. (a) Model selected by `auto.arima`. (b) Holt-Winters parameters.

The automated procedure proposes an ARIMA(1,0,1)(2,1,0)[12] with drift, a distinct specification that omits the regular difference and represents the trend using a drift term and an autoregressive coefficient of 0.969 , very close to unity. This is an alternative parameterization with similar behavior, but clearly inferior in terms of fit: its AIC is -124.15 and its BIC is -105.85 , compared to -146.51 and -137.38 for the manually identified model, and it also uses five parameters instead of two. The analysis of the ACF and PACF thus leads to a solution that is simultaneously more parsimonious and better fitted.

The Holt-Winters model estimates low smoothing parameters for level (0.058) and trend (0.039) and moderate for seasonality (0.196), reflecting a slow evolution of the components, consistent with what was observed in the decomposition.

3.6 Predictive Validation

The three models were used to predict the 24 months of the test set, which were not used in the estimation. The manually identified SARIMA model obtained the best results across all metrics, with an RMSE of 2.647 $\mu\text{g}/\text{m}^3$, a mean absolute error of 2.051, and a mean absolute percentage error of 3.95%, compared to RMSEs of 2.987 and 3.024 for `auto.arima` and Holt-Winters, respectively. Based on a mean level of around 18 $\mu\text{g}/\text{m}^3$, the typical error is approximately 2 $\mu\text{g}/\text{m}^3$ over a two-year horizon, confirming the model's generalizability.

	RMSE	MAE	MAPE
SARIMA manual	0.1362	0.1115	3.9525
auto.arima	0.1518	0.1238	4.3799
Holt-Winters	0.1457	0.1162	3.9716

	modelo	RMSE	MAE
1	SARIMA manual	2.647	2.051
2	auto.arima	2.987	2.247
3	Holt-Winters	3.024	2.223

Figure 15. (a) Accuracy on a logarithmic scale. (b) Accuracy in $\mu\text{g}/\text{m}^3$.

3.7 Waste diagnosis

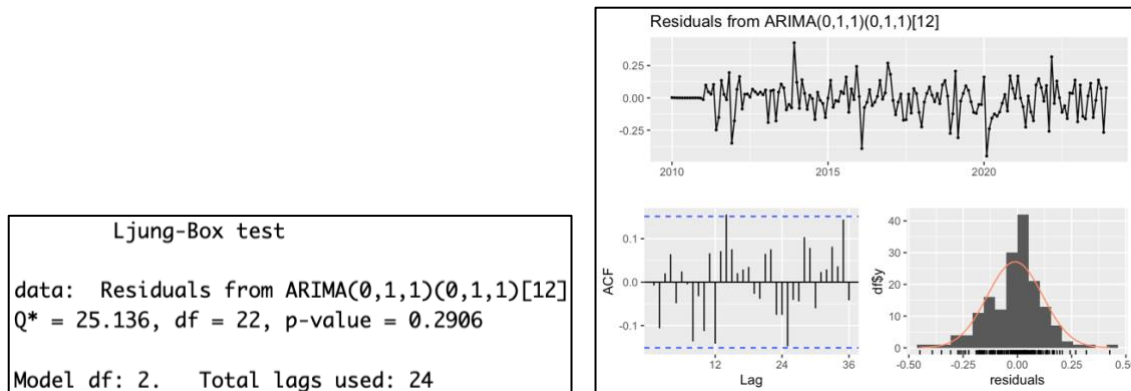


Figure 16. (a) Ljung-Box test. (b) Graphical diagnosis of the residuals.

The Ljung-Box test on 24 lags yielded a statistic of 25.14 with 22 degrees of freedom and a p-value of 0.291, therefore the null hypothesis of no autocorrelation is not rejected. The ACF of the residuals showed no significant coefficients. Consequently, the residuals behaved as white noise, a fundamental condition for the validity of the model.

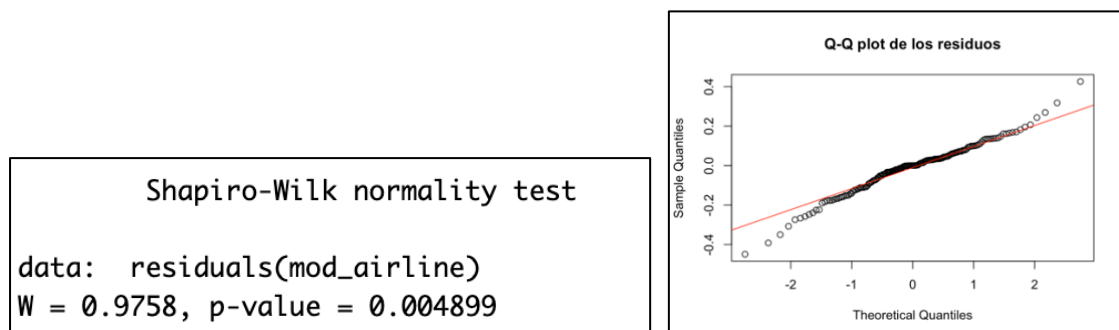


Figure 17. (a) Shapiro-Wilk test. (b) QQ plot of the residuals.

The normality test, however, is rejected ($W = 0.976$; $p = 0.0049$). The QQ plot reveals that the central part of the distribution fits the curve correctly, while both tails deviate from the theoretical line, indicating the presence of extreme values more frequently than expected under normality. This result does not invalidate the coefficient estimates, which remain consistent, but it does affect the width of the prediction intervals, which assume normality. This issue is addressed in the limitations section.

3.8 Final model and prediction

Once the assumptions were verified, the model was re-estimated on the 192 available observations in order to take advantage of all the information for the prediction.

```
Series: log_no2
ARIMA(0,1,1)(0,1,1)[12]

Coefficients:
            ma1      sma1
        -0.9026  -1.0000
s.e.      0.0308   0.1276

sigma^2 = 0.01789: log likelihood = 89.35
AIC=-172.71   AICc=-172.57   BIC=-163.15
```

Figure 18. Final estimated model based on the complete series.

The coefficients remain virtually unchanged compared to those obtained with the training set (-0.903 and -1.000 , versus -0.896 and -1.000), which demonstrates the stability of the model with the addition of two years of data.

```
> tabla_pred
  mes  pred li_95 ls_95
1  Jan 20.36 15.54 26.68
2  Feb 21.93 16.71 28.78
3  Mar 20.33 15.48 26.72
4  Apr 15.40 11.71 20.26
5  May 12.18  9.25 16.05
6  Jun 11.68  8.86 15.41
7  Jul 11.27  8.53 14.88
8  Aug 12.39  9.37 16.38
9  Sep 15.23 11.50 20.16
10 Oct 17.58 13.26 23.30
11 Nov 19.21 14.48 25.50
12 Dec 20.14 15.16 26.76
```

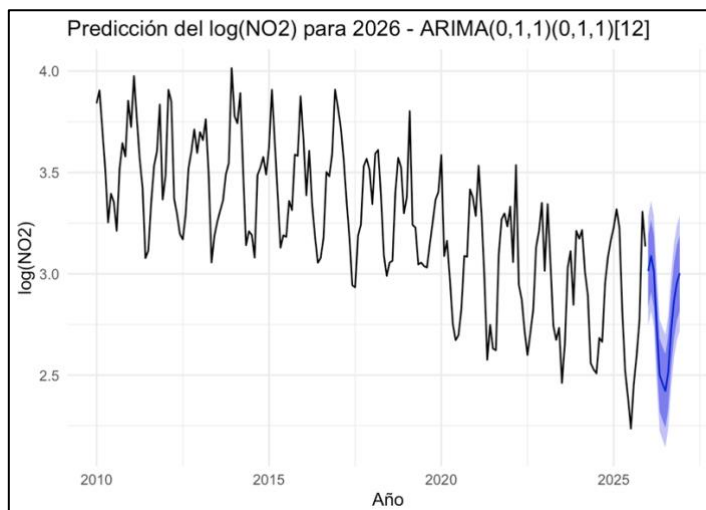


Figure 19. (a) Monthly forecast for 2026. (b) Graphical representation.

The forecast for 2026 places the annual average at $16.5 \mu\text{g}/\text{m}^3$, 9.5% lower than the $18.2 \mu\text{g}/\text{m}^3$ recorded in 2025, and maintains the characteristic seasonal pattern, with a minimum in July of $11.3 \mu\text{g}/\text{m}^3$ and a maximum in February of $21.9 \mu\text{g}/\text{m}^3$. The 95% confidence intervals are wide: in February they range from 16.7 to $28.8 \mu\text{g}/\text{m}^3$, approximately $\pm 30\%$ around the point forecast, reflecting the inherent uncertainty of a twelve-month horizon and the unmodelable meteorological variability. It should be

noted that, after undoing the logarithmic transformation, the values obtained correspond to the conditional median and not the mean, and therefore constitute a slightly conservative estimate.

4. Conclusions

4.1 Analysis Results

The analysis has made it possible to meet the five objectives set out in the introduction.

The NO₂ series from the Stampfenbachstrasse station shows a **sustained downward trend** : the annual average falls from 36.25 µg/m³ in 2010 to 18.22 in 2025, practically half in sixteen years. The decomposition places the decline between 36.66 and 17.96 µg/m³ and allows us to distinguish three phases, with a particularly sharp drop between 2017 and 2020 and a subsequent stabilization around 18-19 µg/m³. This trajectory is consistent with the gradual renewal of the vehicle fleet and the tightening of emissions regulations.

Seasonality is marked and very stable over time. Seasonal indices place the peak in February, 31% above the trend, and the trough in July, 30% below, meaning that the winter level is almost double the summer level. This pattern is attributable to temperature inversions, which limit the vertical dispersion of pollutants during the colder months .

The correlation of 0.883 between the annual level and its variability determined the **multiplicative nature** of the series and, consequently, the application of a logarithmic transformation prior to modeling. This decision, made in the exploratory phase, conditioned all subsequent analysis and proved correct, as it stabilized the variance and allowed working with an additive scheme.

Autocorrelation analysis identified an **ARIMA(0,1,1)(0,1,1)[12]** model, the airline model, which proved to be the best of the three candidates by AIC and BIC. Its validation yields robust results: the residuals behave as white noise according to the Ljung-Box test ($p = 0.291$) and the prediction over two years not used in the fitting reaches a mean absolute percentage error of 3.95%, equivalent to about 2 µg/m³.

One methodological result is worth highlighting. The automated selection procedure proposed a different specification, with five parameters instead of two, which proved inferior in both information criteria and predictive accuracy. Identification based on ACF and PACF led to a solution that was simultaneously simpler and more accurate, illustrating that automated methods are a tool for comparison and not a substitute for analysis.

The **forecast for 2026** places the annual average at 16.5 µg/m³, 9.5% lower than the previous year, maintaining the seasonal pattern. If this forecast holds true, the series would continue to move away from the limits established by the LRV (Limitation Risk Reduction).

4.2 Limitations

The first limitation concerns the **scope of the data**. The analysis is based on a single monitoring station, so the conclusions cannot be extrapolated to the city as a whole or to stations with a different traffic exposure profile. Furthermore, the monthly aggregation, necessary for modeling, eliminates information on short-term pollution episodes, which are precisely the relevant ones from a health and regulatory perspective.

Secondly, the model is **purely univariate**. Based solely on the historical data of the series itself, it does not incorporate meteorological variables (temperature, wind speed, precipitation) or traffic indicators that explain much of the uncaptured variability. This limitation is the trade-off for its simplicity and explains why the prediction is limited to projecting past behavior.

The **rejection of the normality** of the residuals constitutes a third limitation. Although it does not affect the consistency of the estimates, it does mean that the prediction intervals, constructed under the assumption of normality, may underestimate the probability of extreme values. Winter episodes of temperature inversion, which generate occasional peaks, are therefore insufficiently represented in the reported uncertainty.

It should also be noted that the seasonal moving average coefficient is estimated at the **invertibility limit**, indicating a possible overdifferentiation of the seasonal component. In practice, this means that the annual pattern is almost deterministic, a circumstance that the model adequately represents but which suggests that alternatives such as STL decomposition could be equally appropriate.

Finally, the analyzed period includes **the 2020 pandemic**, during which the months of March through July show abnormally low values. The model treats these as ordinary observations, without any specific treatment, a decision justified by the fact that the series subsequently maintained its downward trajectory without any apparent structural break. An intervention analysis would allow for the explicit quantification of the effect.

Three lines of future work should be noted: the extension to a model with exogenous variables (ARIMAX) that incorporates meteorological information, the comparison of the four stations in the city to verify if the trend is homogeneous in the territory, and the use of prediction intervals obtained by resampling, which do not require the assumption of normality.

Annex A. R Code

```
# =====
# AG2 - Time Series | Advanced Statistics (VIU)
# Author: Javier Sarmiento Sanjuán
# Data: Open Data Zürich - Luftqualitätsmessungen (UGZ)
# =====

# ---- 0. Bookstores ----
install.packages(c("forecast", "tseries", "ggplot2", "dplyr", "lubridate"))

library(forecast)
library(tseries)
library(ggplot2)
library(dplyr)
library(lubridate)

# ---- 1. Obtaining the data ----
url_2024 <- "https://data.stadt-zuerich.ch/dataset/ugz_luftschadstoffmessung_tageswerte/download/ugz_ogd_air_d1_2024.csv"
try <- read.csv(url_2024, stringsAsFactors = FALSE, encoding = "UTF-8")

str(test)
head(test)
unique(test$Standort)
unique(test$Parameter)

base_url <- "https://data.stadt-zuerich.ch/dataset/ugz_luftschadstoffmessung_tageswerte/download/ugz_ogd_air_d1_%d.csv"
years <- 2010:2025

read_year <- function(a) {
  message("Downloading ", a)
  df <- read.csv(sprintf(base_url, a), colClasses = "character", encoding = "UTF-8")
  df$Wert <- as.numeric(df$Wert)
  df
}

raw_data <- bind_rows(lapply(anios, read_anio))

dim(datos_raw)
table(substr(raw_data$Datum, 1, 4))

# ---- 2. Data Preparation and Quality ----
no2 <- datos_raw %>%
  filter(Standort == "Zch_Stampfenbachstrasse", Parameter == "NO2") %>%
  mutate(date = as.Date(substr(Datum, 1, 10))) %>%
  arrange(date)

nrow(no2)
range(no2$fecha)
sum(is.na(no2$Wert))
sum(duplicated(no2$fecha))
table(no2$Status)
summary(no2$Wert)

no2 %>% filter(is.na(Wert)) %>% count(month = format(date, "%Y-%m"))

no2_mensual <- no2 %>%
  mutate(anio = year(fecha), mes = month(fecha)) %>%
  group_by(year, month) %>%
  summarise(no2_mean = mean(Wert, na.rm = TRUE),
    n_dias = sum(!is.na(Wert)),
    .groups = "drop") %>%
```

```

arrange(year, month)

nrow(no2_mensual)
summary(no2_mensual$n_dias)
head(no2_mensual)

write.csv(no2_monthly, "no2_zurich_stampfenbach_monthly.csv", row.names = FALSE)

no2_ts <- ts(no2_monthly$no2_average, start = c(2010, 1), frequency = 12)
no2_ts

autoplot(no2_ts) +
  labs(title = "monthly average NO2 - Zürich Stampfenbachstrasse (2010-2025)",
        x = "Year", y = "NO2 (µg/m³)") +
  theme_minimal()

# ---- 3. Exploratory Analysis ----
summary(no2_ts)
sd(no2_ts)
sd(no2_ts) / mean(no2_ts)
round(aggregate(no2_ts, FUN = mean), 2)

boxplot(no2_ts ~ cycle(no2_ts), names = month.abb,
        xlab = "Month", ylab = "NO2 (µg/m³)",
        main = "Distribution of NO2 by month (2010-2025)")

ggseasonplot(no2_ts, year.labels = TRUE, year.labels.left = TRUE) +
  labs(title = "Seasonal pattern of NO2 by year",
        x = "Month", y = "NO2 (µg/m³)") +
  theme_minimal()

# ---- 4. Decomposition of the series ----
annual_means <- as.numeric(aggregate(no2_ts, FUN = mean))
annual_sd <- as.numeric(aggregate(no2_ts, FUN = sd))
round(sd_anuales, 2)
cor(annual_medias, annual_sd)

desc_mult <- decompose(no2_ts, type = "multiplicative")

round(desc_mult$figure, 3)
range(desc_mult$trend, na.rm = TRUE)
sd(desc_mult$random, na.rm = TRUE)

autoplot(desc_mult) +
  labs(title = "Multiplicative decomposition of monthly NO2",
        x = "Year") +
  theme_minimal()

# ---- 5. Transformation and Stationarity ----
log_no2 <- log(no2_ts)

autoplot(log_no2) +
  labs(title = "Logarithm of the mean monthly NO2",
        x = "Year", y = "log(NO2)") +
  theme_minimal()

adf.test(log_no2)
kpss.test(log_no2, null = "Level")

ndiffs(log_no2)
nsdiffs(log_no2)

log_diff <- diff(diff(log_no2, lag = 12), lag = 1)

autoplot(log_diff) +
  labs(title = "Log-transformed series after regular and seasonal differentiation",

```

```

x = "Year", y = "diff(diff(log(NO2), 12))" +
  theme_minimal()

adf.test(log_diff)
kpss.test(log_diff, null = "Level")
length(log_diff)

# ---- 6. Autocorrelation Analysis ----
par(mfrow = c(1, 1))
acf(log_diff, lag.max = 48, main = "ACF of the differentiated series")
pacf(log_diff, lag.max = 48, main = "PACF of the differentiated series")

round(acf(log_diff, lag.max = 24, plot = FALSE)$acf[1:25], 3)
round(pacf(log_diff, lag.max = 24, plot = FALSE)$acf[1:24], 3)

# ---- 7. Model Estimation and Selection ----
train <- window(log_no2, end = c(2023, 12))
test <- window(log_no2, start = c(2024, 1))
length(train); length(test)

mod_airline <- Arima(train, order = c(0,1,1), seasonal = c(0,1,1))
weapon_mod <- Arima(train, order = c(1,1,1), seasonal = c(0,1,1))
mod_ma2 <- Arima(train, order = c(0,1,2), seasonal = c(0,1,1))

data.frame(
  model = c("(0,1,1)(0,1,1)", "(1,1,1)(0,1,1)", "(0,1,2)(0,1,1)"),
  AIC = round(c(mod_airline$aic, mod_weapon$aic, mod_ma2$aic), 2),
  BIC = round(c(mod_airline$bic, mod_weapon$bic, mod_ma2$bic), 2)
)

mod_airline

# ---- 8. Comparison with alternative methods ----
mod_auto <- auto.arima(train, seasonal = TRUE, stepwise = FALSE,
  approximation = FALSE)
mod_auto

mod_hw <- HoltWinters(train, seasonal = "additive")
mod_hw$alpha; mod_hw$beta; mod_hw$gamma
mod_hw$coefficients[1:2]

# ---- 9. Predictive validation on the test set ----
pred_airline <- forecast(mod_airline, h = 24)
pred_auto <- forecast(mod_auto, h = 24)
pred_hw <- forecast(mod_hw, h = 24)

acc <- rbind(
  "SARIMA manual" = accuracy(pred_airline, test)[2, c("RMSE", "MAE", "MAPE")],
  "auto.arima" = accuracy(pred_auto, test)[2, c("RMSE", "MAE", "MAPE")],
  "Holt-Winters" = accuracy(pred_hw, test)[2, c("RMSE", "MAE", "MAPE")]
)
round(acc, 4)

test_orig <- window(no2_ts, start = c(2024, 1))
rmse_real <- function(p) sqrt(mean((exp(p$mean) - test_orig)^2))
mae_real <- function(p) mean(abs(exp(p$mean) - test_orig))

data.frame(
  model = c("SARIMA manual", "auto.arima", "Holt-Winters"),
  RMSE = round(sapply(list(pred_airline, pred_auto, pred_hw), rmse_real), 3),
  MAE = round(sapply(list(pred_airline, pred_auto, pred_hw), mae_real), 3)
)

# ---- 10. Waste Diagnosis ----
checkresiduals(mod_airline)

```

```

shapiro.test(residuals(mod_airline))

qqnorm(residuals(mod_airline), main = "QQ plot of residuals")
qqline(residuals(mod_airline), col = "red")

# ---- 11. Final model and prediction for 2026 ----
mod_final <- Arima(log_no2, order = c(0,1,1), seasonal = c(0,1,1))
mod_final

pred_2026 <- forecast(mod_final, h = 12, level = c(80, 95))

table_pred <- data.frame(
  month = month.abb,
  pred = round(exp(pred_2026$mean), 2),
  li_95 = round(exp(pred_2026$lower[, 2]), 2),
  ls_95 = round(exp(pred_2026$upper[, 2]), 2)
)
table_pred

round(mean(exp(pred_2026$mean)), 2)
round(mean(window(no2_ts, start = c(2025, 1))), 2)

autoplot(pred_2026) +
  labs(title = "Prediction of log(NO2) for 2026 - ARIMA(0,1,1)(0,1,1)[12]",
    x = "Year", y = "log(NO2)") +
  theme_minimal()

```

Annex B. Data Source

The data are from the *Täglich aktualisierte Luftqualitätsmessungen, seit 1983 dataset*, published by the Umwelt- und Gesundheitsschutz Zürich (UGZ) under a Creative Commons CC0 license, and are available in annual CSV files at:

https://data.stadt-zuerich.ch/dataset/ugz_luftschadstoffmessung_tageswerte

The download and processing were performed directly from R, according to the code collected in Annex A.